

VINAYAKA SAJJANSHETTY

R&D Engineer | Robotics & AI Prototyping |

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PROFESSIONAL SUMMARY

Research & Technical Engineer with a strong background in industrial IoT, automation, and data-driven systems. Experienced in applied research, product validation, and system reliability improvement within manufacturing environments. Proven ability to investigate complex technical issues, translate findings into concrete software and system changes, and collaborate with cross-functional and international teams (US–Japan) to deliver stable, scalable, and production-ready solutions for factory operations.

EXPERIENCE

Somic Ishikawa

Research & Technical Engineer

Japan, JP
10/2024 - Present

Leading initiatives in manufacturing automation, specializing in robotics R&D, integration of emerging technologies, and international project coordination.

R&D Engineer → Product-Focused Role

- Led on-site validation of an electrical power testing machine for factory operators and manufacturing teams, enabling real-time voltage and power analysis over Wi-Fi.
- Owined investigation of intermittent zero-data transmission issues that blocked downstream analytics and ML-based efficiency calculations.
- Designed stress test cycles (increasing node counts) to isolate routing failures, identifying limits in the shortest-path algorithm.
- Drove a software logic update to dynamically reset shortest-path parameters, stabilizing packet transmission and restoring data reliability.
- Served as primary EN–JP coordination point between US and Japan teams, reducing test-cycle misunderstandings and accelerating product acceptance.

Robotic Hand R&D (Internal Project)

- Explored an automated robotic hand concept for factory managers to improve efficiency by automating repeatable assembly work.
- Defined usage scenarios (standing and moving work) to evaluate feasibility of consistent, human-like motion in production settings.
- Validated first-phase feasibility through successful standing-work cycles (left/right), demonstrating repeatable motion reliability.
- Identified finger-grip limitations during moving tasks, defining clear next-step requirements toward a future product roadmap.

OPC UA Communication Transformation (Platform / Infrastructure Product)

- Contributed to an internal IoT initiative to modernize factory communication by migrating legacy protocols to a scalable OPC UA–based architecture.
- Enabled real-time data exchange by integrating PLC data via a custom FINS protocol handler (Python) into an OPC UA server.
- Delivered real-time visualization through Grafana dashboards and configured Prometheus alerts to monitor system performance.
- Improved machine-to-system communication, real-time monitoring, and operational transparency while reducing data handling complexity for factory teams.

Legato Health Technologies

Software Engineer- Data & Analytics

Bangalore, India, IN
09/2022 - 09/2024

- Supported a healthcare analytics product used by business and operations teams to manage and analyze clinical, claims, insurance, and billing data.
- Collaborated with business stakeholders, QA, and operations to understand new data requirements and ensure changes did not impact existing database entities.
- Improved data accuracy and reliability by performing root-cause analysis on data issues and supporting migration debugging.
- Reduced issue resolution time and improved delivery predictability by streamlining communication across development, QA, and operations teams.

Bosch Automotive Electronics Pvt. Ltd.

Engineering Intern

Bangalore, India, IN
01/2019 - 03/2019

- Analyzed heat-induced failures in the selective soldering process that reduced filter assembly reliability and increased chip wastage.
- Proposed and built three POC monitoring models, including a mini-camera–based setup and time-based usage tracking to study operating conditions and degradation patterns.
- Enabled data-driven identification of optimal operating windows and maintenance timing, supporting reduced material wastage and improved production efficiency.

EDUCATION

Shizuoka University, Japan JP 10/2022 - 09/2024
(Master of Science in Computer Science)

Relevant Course work: X-ray Backscattering, Photonics, Image Processing, Probability & Discrete Mathematics, Machine Learning

PES University, Bangalore, India IN

(B.E Electronics Communication Eng) 07/2015 - 06/2019

KEY ACHIEVEMENTS

- Enabled international product validation by bridging technical and bilingual (EN–JP) communication between US and Japan teams, accelerating issue resolution and product acceptance.
- Validated feasibility of a robotic hand automation concept through phased internal demos, demonstrating repeatable standing-work motions and defining next-step requirements for future product development.
- Developed and delivered a FINS–OPC UA integration module in Python, enabling seamless PLC data access and standardized data exchange across systems.

Volunteering & Teaching

NPO Volunteer – Cross-Cultural Exchange (Japan)

Organized and delivered cultural presentations on India, promoting cross-cultural understanding and community engagement within local Japanese groups.

Part-Time English Instructor

Taught practical English to high school students and adult learners preparing for international travel, focusing on real-world communication and fluency.

LANGUAGES

English ●●●●●●
Japanese ●●●●●●
Hindi ●●●●●●

SKILLS

Monitoring, Visualization & Platform Observability

Real-time dashboards and visualization: Grafana, Plotly
System monitoring & alerting: Prometheus
ETL pipelines and real-time data flow design

Data, Analytics & Applied Machine Learning

Data analysis & modeling: pandas, NumPy, scikit-learn
Image processing & computer vision: OpenCV
ML prototyping and experimentation TensorFlow, Pytorch

Industrial IoT, Robotics & Embedded Systems

Industrial communication: OPC UA, PLC data integration
Embedded platforms: Raspberry Pi, Jetson Nano
Robotics frameworks: ROS2, RViz2, Gazebo
Sensor and system-level integration for factory environments

Programming & Systems Engineering

Python (production scripting, data pipelines, protocol integration)
C / C++ (embedded and performance-critical components)
Linux, Git, GitHub, Docker